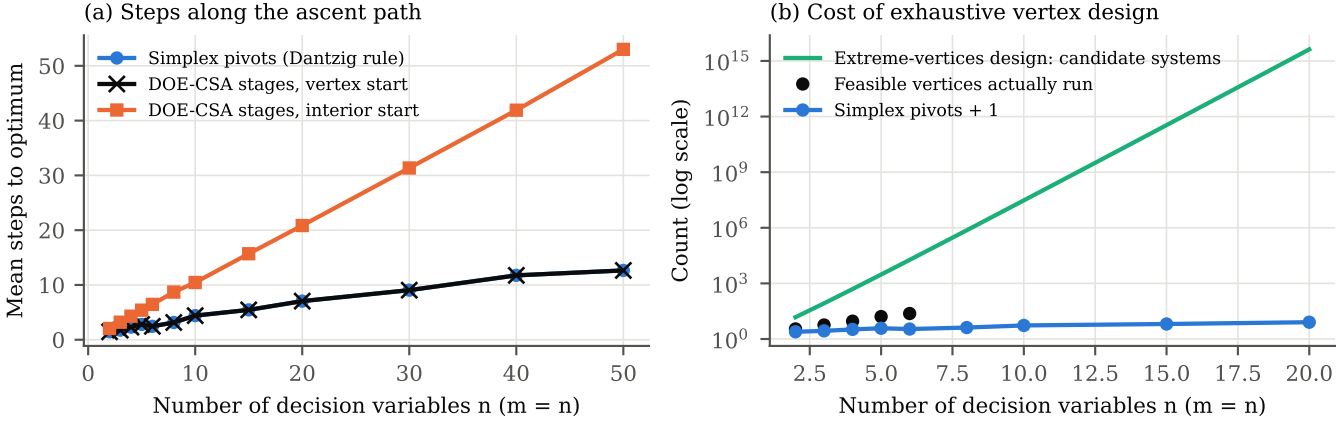
(a) Steps along the ascent path
Simplex pivots (Dantzig rule)
DOE-CSA stages, vertex start
DOE-CSA stages, interior start
Mean steps to optimum
0
10
20
30
40
50
Number of decision variables n (m = n)
(b) Cost of exhaustive vertex design
Extreme-vertices design: candidate systems
Feasible vertices actually run
Simplex pivots + 1
Count (log scale)
10^{0}
10^{3}
10^{6}
10^{9}
10^{12}
10^{15}
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Number of decision variables n (m = n)